

Gene Pulser® Electroprotocols

* We recommend adapting this protocol to use the Gene Pulser electroporation buffer (catalog #165-2676, 165-2677), which increases cell viability and transfection efficiency in mammalian cell lines.

Cell Type	Mammalian, adherent	Molecules Electroporated	DNA: neo gene.
Species Used	Human, MRC-5 / V1, lung fibroblasts, transformed		

Before the Pulse

Cell Growth Medium	DMEM + 10% Fetal Calf Serum (GIBCO/BRL, Sigma)	Growth Phase at Harvest	Log phase
		Pre-pulse Incubation	37° C, 1/2 hour
Wash Solution	Phosphate Buffered Saline		

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature	25 °C	Cuvette Gap	0.4 cm
Electroporation Medium*	DMEM	Voltage	2.0 kV
Cell Density	5 x 10 (6) cells	Field Strength	5.0 kV/cm
Volume of Cells	1.4 ml (**see notes)	Capacitor	25 µF
DNA Concentration	20 µg	Resistor	(Pulse Controller) Ω none; NOT
DNA Resuspension Buffer	water	Time Constant	0.4 msec
Volume of DNA	20 µl		
After the Pulse			
Outgrowth Medium	DMEM, 10% Fetal Calf Serum		

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature	37 °C
Length of Incubation	1 month for selection
Selection Method or Assay Used	1 mg / ml geneticin
Electroporation Efficiency	Not given
Per Cent Survival	Not given

**It is NOT RECOMMENDED to use high voltage with out the Pulse Controller. It is not recommended to use more than 0.8 ml in the 0.4 cm cuvette; greater volumes may create non-uniform field strengths during the pulse.

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Survey Number

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